Enforced Abstention Prevents Silent Errors in Biomedical Evidence Pipelines

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Abstract

Pipelines that assemble public biomedical records into scientific claims are evaluated on what they emit. We show that this measurement misses their dominant failure mode. A *silent error* is an output that is wrong, well-formed, free of any exception, and indistinguishable downstream from a correct result. We introduce *silent-error ablation*, a protocol that removes a pipeline's verification guards and counts what the unguarded pipeline then emits over identical inputs. We apply it to a rare-disease drug repurposing pipeline in which every stage verifies its input against a stated precondition or declines it, and every decline is counted as an output. Over 746 clinical variant records for 8 genes, removing residue verification makes 16 of 114 emitted protein sequences silently wrong (14.0%). Removing copy-number exclusion attributes 382 of 746 records to genes they say nothing about, moving 3 of 8 genes from apparent support to zero. We report the cost alongside the benefit: the sequence lane declines 266 of 364 records. We also report two defects the protocol found in our own instrumentation, including a rate we had overstated by ten percentage points. Both had the failure shape the protocol targets, and a mechanical check was what exposed each one.

1 Introduction

Computational drug repurposing produces rankings. Recent systems score millions of drug–disease pairs [Huang et al., 2024], and commercial databases sell curated competitive landscapes. Evaluation of these systems concentrates on the quality of what they emit. They rarely record which evidence produced a given score, and they rarely offer a principled way to report that the evidence was insufficient, so a confident guess and an auditable finding reach the reader in the same form.

The consequence matters because of how these pipelines fail. Their dominant failure mode is the *silent error*: an output that is (i) wrong, (ii) syntactically well-formed, (iii) free of any exception or warning, and (iv) indistinguishable by a downstream consumer from a correct output. Silent errors are the non-statistical analogue of hallucination, plausible and confident at the point of use. They are also invisible to any evaluation that scores only the outputs a pipeline chooses to produce, since by construction the pipeline believes them.

Our claim concerns the refusals. A pipeline can be built so that each stage verifies its input against a stated precondition or declines it, with every decline counted and given a reason, and this discipline prevents a measurable class of silent error. We make three contributions.

1. **Silent-error ablation**, an evaluation protocol for evidence pipelines. We score the pipeline's refusals by removing each verification guard and counting what the resulting pipeline emits over identical inputs. Scoring its recommendations instead would require wet-lab validation, which we have not performed.

2. A **verify-or-abstain architecture** in which a small set of readable rules assigns every verdict, model output is excluded from those decisions, and every record is content-addressed under a single Merkle root that a reader re-derives client-side.

3. **Measurements** on 746 real public records, reported with their cost and their limits, including two defects the protocol exposed in our own instrumentation.

2 Related work

Repurposing at scale. Graph-based models predict drug–disease associations across large biomedical knowledge graphs [Huang et al., 2024]. These systems optimise coverage and accuracy. Provenance and calibrated refusal fall outside their objective, and their outputs are designed for use in place of independent re-derivation.

Selective prediction. The option to abstain has a long history in classification, from the reject-option formulation [Chow, 1970] through selective classification with coverage guarantees [El-Yaniv and Wiener, 2010, Geifman and El-Yaniv, 2017]. That literature triggers abstention from model confidence. Our abstentions follow from a failed precondition on the input record, which makes them checkable, deterministic, and reportable with a reason.

Provenance and tamper-evidence. We adopt the Merkle audit-tree construction of Certificate Transparency [Laurie et al., 2013], which supplies inclusion proofs and exposes any post-hoc edit.

Missing data. Rubin's taxonomy [Rubin, 1976] describes the defect we found in our own acquisition stage. The loss was *missing not at random*, since the probability of a record being dropped followed from its own size, which in turn followed from its class.

3 Approach

3.1 One rule

The pipeline rests on a single constraint: **every stage must either verify its input against a stated precondition, or decline it**. A decline is itself an output. It carries a machine-readable reason and enters the same ledger as an admitted record.

The commitment is stronger than it sounds, because it removes the fallback paths a pipeline usually relies on. A symbol that fails to resolve stays unresolved, with approximate matching disallowed. A record whose numbering disagrees with its reference is set aside. A source that a third party would be unable to re-query is left out of the system entirely.

The input is a rare-disease name. The output is a set of (disease, target, drug-programme) rows. Each row carries a verdict in {GAP, NOT_A_GAP, ABSTAIN}, the identifier of the rule that produced it, and a receipt: the content digests of every record the verdict consumed.

3.2 Six types of evidence

We collect six kinds of evidence, each answering exactly one question (Table 1). The separation is deliberate. Each source is consulted for the one thing it is authoritative about, so that a failure in one lane stays confined to that lane.

Two details matter for the evaluation that follows. First, a trial retains its *status and phase* alongside the fact of its existence. A terminated Phase 3 is evidence against a pairing, and a pipeline that records only existence converts that past failure into an apparent opportunity.

Second, the canonical sequence enters the system as evidence in its own right. It is the frame against which every variant record is checked, and the first guard we ablate is precisely a check against it.

Table 1: The six evidence types. Each source answers one question, and each is consulted only within its competence.

Evidence type	Question it answers	Admitted as
Target association	which genes are implicated in this disease?	target set
Target tractability	is there existing chemistry against them?	druggability
Clinical variants	do patients carry damaging variants here?	genetic support
Canonical sequence	what is the reference protein?	substitution frame
Registered trials	has this pairing been tried, and how did it end?	prior attempt
Regulatory status	is the drug approved, and for what?	repurposing cost
Cell morphology	does any compound move cells toward reference?	phenotype rank

3.3 The provenance framework

Every response is hashed the moment it arrives, and its query URL is recorded beside the digest. Records carry one of two custody levels.

- **Recomputed.** We hold the bytes, and the digest can be re-derived from them locally.
- **Committed.** We hold a digest captured at origin while the bytes stay with the publisher. This is the honest level for a large third-party artifact we decline to redistribute.

The dashboard renders the distinction, because the two levels support different claims. Every record, at either level, is content-addressed and committed under a single Merkle root, using the audit-tree construction of Certificate Transparency [Laurie et al., 2013]. Verification re-derives the root in the reader's browser from the record bytes. A single-byte edit moves the root, and the first divergent node is named.

Admissibility follows from this. A source whose responses a third party would be unable to re-obtain can carry no meaningful digest, which makes it inadmissible. One candidate source is excluded on exactly this ground, and we report the exclusion and its reason.

3.4 Every record lands somewhere

The provenance framework is half of what the evaluation needs. The other half is a *closed disposition vocabulary*. Each stage assigns every record it sees to exactly one labelled outcome, and the labels are exhaustive.

The variant lane admits a record, excludes it as a multi-gene copy-number event, or reports it unavailable from the source. The sequence lane emits a substitution or a truncation, or declines with one of six named reasons (Table 4). The decision layer assigns one of five verdicts through nine first-match rules. Every path through every stage ends at a label.

Each stage then asserts its own arithmetic: records in must equal records out plus records declined. For the corpus used here, acquisition reconciles as 746 fetched = 364 admitted + 382 excluded + 0 unavailable. The sequence lane reconciles as 364 in = 98 emitted + 266 declined.

3.5 Provenance makes the ablation measurable

Custody is usually motivated by trust: a sceptical reader can check the evidence for themselves. That holds here, and a second property is what makes the framework load-bearing in this paper.

The ablation is measurable only because the pipeline is fully accounted. To report what an unguarded pipeline *would have emitted*, we need the complete set of records it would have seen and the disposition each one actually received. A pipeline that drops records on an unlabelled path leaves its own denominator unknown, and careful reading of the code recovers none of it.

We observed this directly. Section 5 reports that our own acquisition stage was discarding 100 records through an unlabelled branch. The per-gene counts looked entirely normal, and review had passed over the gap that the reconciliation assertion then exposed. Until it was closed, every rate in this paper had an unknown denominator.

3.6 Decision

Nine first-match rules assign the verdict. They cover unresolved symbols, absent targets, absent programmes, failed lookups, prior trials, and three grades of untried pairing. The rules are plain code and are read in order. Setting a verdict, resolving a symbol, and choosing a threshold are all closed to model output. A language model is used only to phrase rows whose verdict is already fixed.

Identity resolution deserves its own note, because it is where silent errors originate. Symbols resolve by exact match or by an explicit alias table, and those two paths are the only ones available. Collisions are real. One symbol in our corpus denotes two unrelated proteins, and a similarity match would return a valid gene, succeed at every downstream step, and answer a question about the wrong molecule.

3.7 The guards under test

Two guards carry the measurements in Section 4.

G1, residue verification. A variant record names a wild-type residue and a position. Before applying the substitution to the canonical sequence, the pipeline checks that the named residue is what the canonical sequence actually holds at that position. A disagreement means the record is numbered against a different transcript, and the row declines.

G2, copy-number exclusion. A variant record is admitted as evidence about a gene only when it is gene-specific. Records typed as copy-number events, or spanning more than a small number of genes, are excluded. A deletion covering hundreds of genes is evidence about the interval and silent about each gene inside it.

4 Evaluation

4.1 Setup and metric

We compare two pipelines over identical inputs. **Enforcement on** is the system as described. **Enforcement off** is the same system with one guard removed, which is the implementation obtained by following each source's documentation and anticipating no specific failure. Every other guard stays active in both conditions.

We report the *silent error rate*: the fraction of outputs the unguarded pipeline emits that are demonstrably wrong. Loud failures, such as a crash or an out-of-range index, fall outside the numerator, since a downstream consumer sees them arrive. Absolute counts accompany every rate, because the denominators here are small enough that a percentage alone would misrepresent them.

The corpus consists of pathogenic and likely-pathogenic clinical variant records for 8 genes, captured 2026-08-25, with subset digest `3aa5e6723563`. Acquisition reconciles: 746 records fetched = 364 admitted + 382 excluded + 0 unavailable.

4.2 Results

Table 2 gives the headline result. Removing G1 makes 16 of 114 emitted sequences wrong. Each one is valid FASTA built from real residues, and a structure-prediction tool accepts it without complaint and returns a structure for a protein that has never existed. With G1 active the lane

Table 2: Silent errors prevented. Rates are over what the unguarded pipeline emits, with loud failures excluded from the numerator.

Guard	Naive emits	Silently wrong	Rate	Enforced emits
G1 residue verification	114	16	14.0%	98
G2 copy-number exclusion	746	382	51.2%	364

emits 98 sequences, 40 substitutions and 58 truncations, and we verified all 98 of them position by position against their wild-type reference.

Removing G2 attributes 382 of 746 records to genes they say nothing about. The per-gene effect (Table 3) varies sharply, and this variation carries the more interesting result. 3 of 8 genes (PGS1, PHB2, CHCHD3) fall from apparent evidentiary support to an honest zero. The unguarded pipeline reports all three as supported, and for one of them the top-ranked supporting evidence is a chromosomal deletion syndrome.

Table 3: Per-gene effect of G2. * marks genes reduced to zero.

Gene	Unfiltered	After G2	Excluded
TAFAZZIN	339	109	230
CRLS1	33	3	30
PGS1*	14	0	14
PTPMT1	13	1	12
HADHA	267	251	16
PHB	0	0	0
PHB2*	46	0	46
CHCHD3*	34	0	34

4.3 The cost of abstention

A trivial policy that declines everything also achieves a zero silent error rate, and it is useless. The guards therefore have to decline *selectively*. The sequence lane declines 266 of 364 records (Table 4). Most declines are records carrying no protein-level change, or frameshifts whose downstream sequence the record leaves undetermined, and a substituting pipeline would emit neither class. Among the records that are genuinely substitution-eligible, the guard declines 16 and emits 98, all verified correct.

Table 4: Why the sequence lane declines. Only *residue mismatch* covers records that a substituting pipeline would otherwise have emitted.

Reason declined	Records
frameshift	131
no protein notation	118
residue mismatch	16
position out of range	1

5 Defects found in our own instrumentation

We applied the protocol to our own pipeline before submission. Two findings bear directly on the results above, and we report them here because leaving them out would reproduce the failure this paper describes.

The acquisition stage discarded records without counting them. A reconciliation check requires fetched records to equal admitted plus excluded plus unavailable. That check revealed 100 records leaving through a single unlogged branch. The cause lay upstream: the source declines to serialise a response above a fixed size, and one record listing over a thousand genes carried its whole batch past that limit. Our error handler read three possible error keys, and the source reports this condition under a fourth, so our own log recorded a generic "no result" string. *We had built a guard that detected the loss and discarded its explanation.* Under Rubin's taxonomy [Rubin, 1976] the loss was *missing not at random*: the records we lost were large, their size followed from spanning many genes, and spanning many genes is exactly the class G2 exists to exclude. Left uncorrected it biased our own G2 rate downward by roughly six percentage points.

Our first reported rate was wrong by ten points. An independent recount disagreed with the pipeline's own count of G1 rejections. The pipeline tested the residue precondition before classifying the variant, so frameshift records that also mismatched were filed as residue mismatches, which inflated the reported figure with records that a substituting pipeline could never emit. The published rate was 24.6%, and the correct rate is 14.0%. The emitted sequences were byte-identical before and after the fix, since the guard had rejected exactly the same records and only the reasons were wrong. The defect lay entirely in the measurement, which is why the system looked healthy and why recounting was the only thing that exposed it.

We draw one narrow conclusion. Mechanical reconciliation, asserted at every boundary, caught defects that careful reading missed, including defects in the instrument that produces this paper's numbers.

6 Limitations

The G1 effect comes from one gene. All 16 residue mismatches occur in a single gene. A second gene contributes more emitted sequences than that one and produces zero. The measured G1 result therefore reflects one transcript-numbering convention, and $n = 1$ describes it more honestly than $n = 8$.

The declined records are recoverable, and we discard them. We tested the failing records directly and found that a single constant offset of +30 residues reconciles all 16 of them exactly. They are correctly numbered against a longer transcript, and they become wrong only on contact with the canonical sequence. Abstention is the safe response here. A pipeline that detected the offset would recover them, and ours throws them away.

Small denominators and a single corpus. The rates come from hundreds of records, drawn from one snapshot of one source. That source is live, and two honest runs a day apart returned different totals, which is why we pin a capture date and publish the digest of the exact table used.

We evaluate our own guards on data we selected. We designed the guards, and the failures they catch are failures we anticipated. This measures the value of enforcement given a correct precondition. It leaves open how many preconditions we failed to think of, and Section 5 shows that this residual exceeds zero.

We do not evaluate the science. Establishing whether any proposed pairing is biologically correct requires wet-lab validation we have not performed. The system's claim ceiling is enforced in code and permits a repurposing hypothesis, stopping short of any therapeutic claim.

7 Conclusion

Provenance and abstention are usually argued from principle. We argue for them from a measurement: on real public records, removing two verification guards causes a pipeline to emit 16 of 114 silently wrong sequences and to misattribute 382 of 746 variant records. The intervention is architectural, so it transfers to any pipeline that assembles evidence into claims, whether or not a language model participates. The most persuasive evidence we can offer is that the discipline caught our own instrument twice, and that a mechanical check was what caught it each time.

Two directions follow. Custody can extend to the turn level of the model interactions that assist the pipeline, so that the assistance itself carries receipts. The transcript-offset records can be recovered in place of being declined.

Reproducibility statement

Every figure in this paper is generated from the pipeline outputs by a script, and no number is typed by hand. A second script scans the prose for any figure that a later run has superseded. An anonymised artifact containing the code, the captured corpus with digests, and a one-command regeneration of every table is available to reviewers [link withheld for anonymous review].

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